Constructing a Simple IPv6 Network

Difficulty (out of 5 stars): ★★★ Recommended study time: 3 hours

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1 About This Lab

1.1 Task Description

An enterprise network remains IPv4-based; however, with technological advancements and transitions, there's a necessity for the network to migrate from IPv4 to IPv6. As an administrator, your initial task involves designing and reconstructing the current network to support IPv6. In this lab, configure IPv6 addresses, IPv6 static routes, and DHCPv6.

After completing this task, you will gain a clear understanding of IPv6 and be able to independently set up and configure a small IPv6 network.

1.2 Objectives

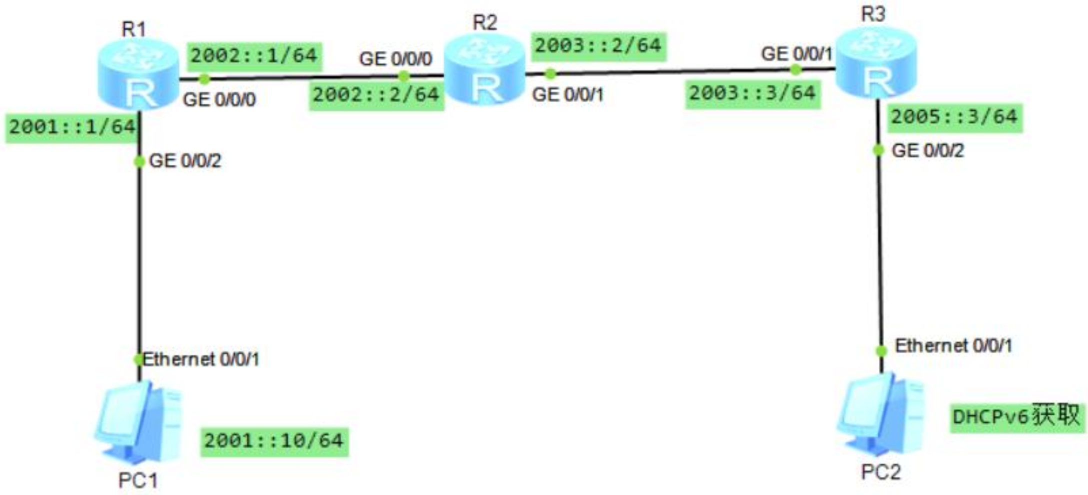
On completing this task, you will be able to:

1. Configure IPv6 addresses.
2. Configure IPv6 static routes.
3. Configure a DHCPv6 server.
4. Be familiar with IPv6 network test methods.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]ipv6
```

```
[R1]
[R1]int gi0/0/0
[R1-GigabitEthernet0/0/0]ipv6 enable
[R1-GigabitEthernet0/0/0]ipv6 address 2002::1/64
[R1-GigabitEthernet0/0/0]quit
[R1]
[R1]int gi0/0/2
[R1-GigabitEthernet0/0/2]ipv6 enable
[R1-GigabitEthernet0/0/2]ipv6 address 2001::1/64
[R1-GigabitEthernet0/0/2]quit
[R1]
```

- Initial configuration of R2

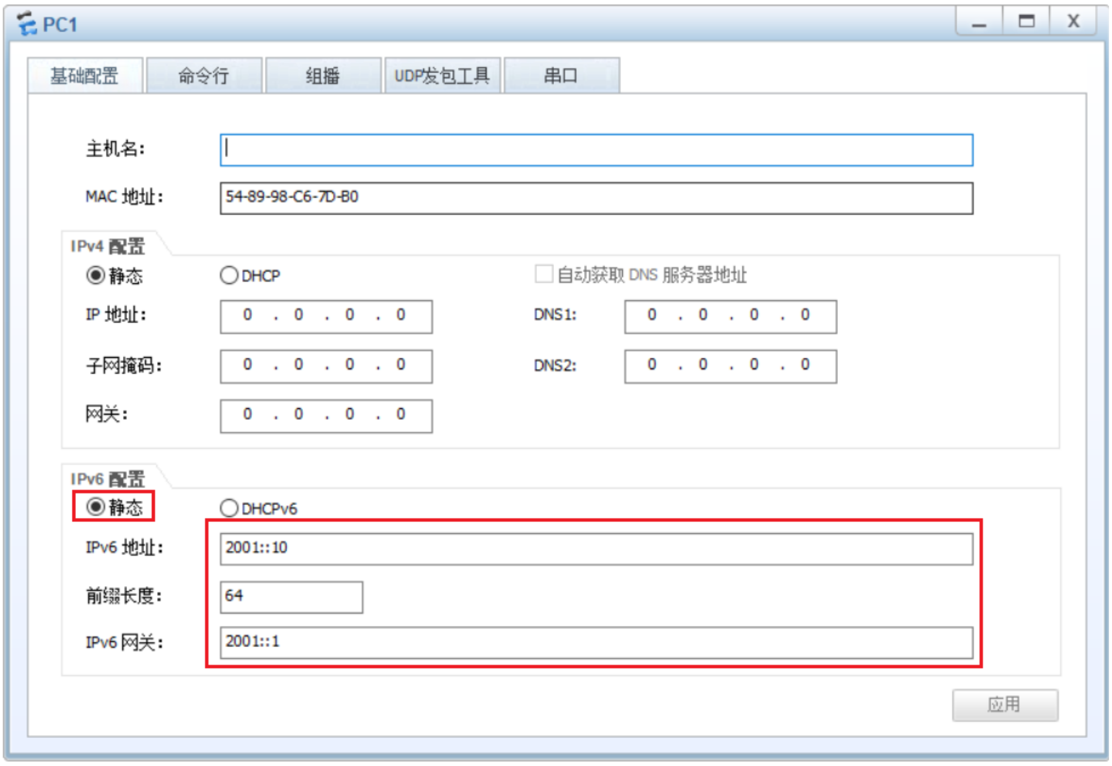
```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]ipv6
[R2]int gi0/0/0
[R2-GigabitEthernet0/0/0]ipv6 enable
[R2-GigabitEthernet0/0/0]ipv6 address 2002::2/64
[R2-GigabitEthernet0/0/0]quit
[R2]int gi0/0/1
[R2-GigabitEthernet0/0/1]ipv6 enable
[R2-GigabitEthernet0/0/1]ipv6 address 2003::2/64
[R2-GigabitEthernet0/0/1]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]ipv6
[R3]int gi0/0/2
[R3-GigabitEthernet0/0/2]ipv6 enable
[R3-GigabitEthernet0/0/2]ipv6 address 2005::3/64
[R3-GigabitEthernet0/0/2]quit
[R3]
[R3]int gi0/0/1
[R3-GigabitEthernet0/0/1]ipv6 enable
[R3-GigabitEthernet0/0/1]ipv6 address 2003::3/64
[R3-GigabitEthernet0/0/1]quit
[R3]
```

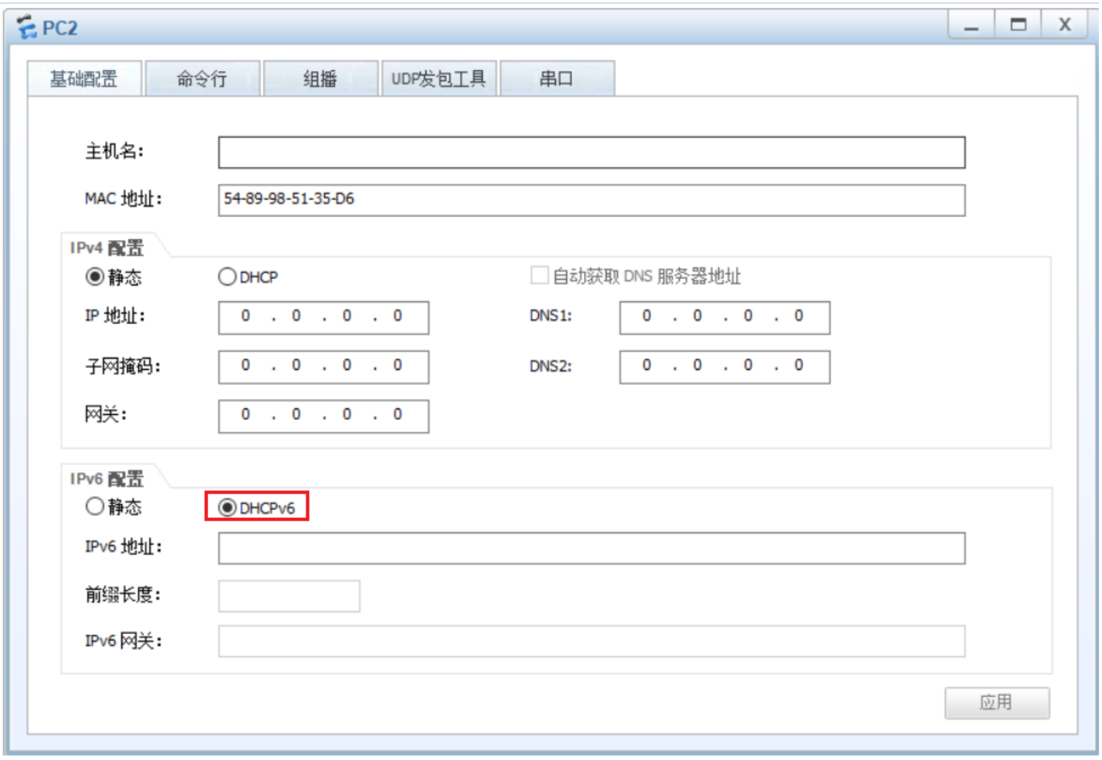
- IPv6 configuration of PC1, as shown in Figure 2

Figure 2-2 IPv6 configuration of PC1



- IPv6 configuration of PC2, as shown in Figure 3

Figure 2-3 IPv6 configuration of PC2



3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ipv6 interface brief** command on R1, R2, R3, R4, and R5 to check interface configurations.

```
<R1>display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                      Physical                Protocol
GigabitEthernet0/0/0           up                      up
[IPv6 Address] 2002::1
GigabitEthernet0/0/2           up                      up
[IPv6 Address] 2001::1
<R1>

<R2>display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                      Physical                Protocol
GigabitEthernet0/0/0           up                      up
```

```
[IPv6 Address] 2002::2
GigabitEthernet0/0/1              up                          up
[IPv6 Address] 2003::2
<R2>

<R3>display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                         Physical                    Protocol
GigabitEthernet0/0/1              up                          up
[IPv6 Address] 2003::3
GigabitEthernet0/0/2              up                          up
[IPv6 Address] 2005::3
<R3>
```

Run the **ping** command on R1 and R2 to check the connectivity between devices.

```
<R1>ping ipv6 -c 2 2001::10
  PING 2001::10 : 56   data bytes, press CTRL_C to break
    Reply from 2001::10
    bytes=56 Sequence=1 hop limit=255   time = 20 ms
    Reply from 2001::10
    bytes=56 Sequence=2 hop limit=255   time = 10 ms

  --- 2001::10 ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 10/15/20 ms

<R2>ping ipv6 -c 2 2002::1
  PING 2002::1 : 56   data bytes, press CTRL_C to break
    Reply from 2002::1
    bytes=56 Sequence=1 hop limit=64   time = 30 ms
    Reply from 2002::1
    bytes=56 Sequence=2 hop limit=64   time = 20 ms

  --- 2002::1 ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/25/30 ms

<R2>ping ipv6 -c 2 2003::3
  PING 2003::3 : 56   data bytes, press CTRL_C to break
    Reply from 2003::3
    bytes=56 Sequence=1 hop limit=64   time = 80 ms
    Reply from 2003::3
    bytes=56 Sequence=2 hop limit=64   time = 30 ms
```

```
 --- 2003::3 ping statistics ---
   2 packet(s) transmitted
   2 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 30/55/80 ms

<R2>
```

3.2 Testing Network Connectivity on PC1

On PC1, ping the IPv6 addresses of R2 and R3.

```
PC>ping 2001::1

Ping 2001::1: 32 data bytes, Press Ctrl_C to break
From 2001::1: bytes=32 seq=1 hop limit=64 time=16 ms
From 2001::1: bytes=32 seq=2 hop limit=64 time=15 ms
From 2001::1: bytes=32 seq=3 hop limit=64 time<1 ms
From 2001::1: bytes=32 seq=4 hop limit=64 time=16 ms
From 2001::1: bytes=32 seq=5 hop limit=64 time=15 ms

--- 2001::1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 0/12/16 ms

PC>
PC>ping 2002::2

Ping 2002::2: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 2002::2 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
  100.00% packet loss

PC>ping 2003::3

Ping 2003::3: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!
```

```
--- 2003::3 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
  100.00% packet loss

PC>
```

The command output shows that PC2 can ping R1's IPv6 address, but cannot ping R2's and R3's IPv6 addresses.

3.3 Configuring IPv6 Static Routes

Configure IPv6 static routes on R1, R2, and R3.

```
[R1]ipv6 route-static 2005:: 64 2002::2
[R1]ipv6 route-static 2003:: 64 2002::2

[R2]ipv6 route-static 2001:: 64 2002::1
[R2]ipv6 route-static 2005:: 64 2003::3

[R3]ipv6 route-static 2001:: 64 2003::2
[R3]ipv6 route-static 2002:: 64 2003::2
```

3.4 Testing Connectivity on PC1 Again

On PC1, ping the IPv6 addresses of R2 and R3.

```
PC>ping 2003::3

Ping 2003::3: 32 data bytes, Press Ctrl_C to break
From 2003::3: bytes=32 seq=1 hop limit=62 time=31 ms
From 2003::3: bytes=32 seq=2 hop limit=62 time=16 ms
From 2003::3: bytes=32 seq=3 hop limit=62 time=15 ms
From 2003::3: bytes=32 seq=4 hop limit=62 time=32 ms
From 2003::3: bytes=32 seq=5 hop limit=62 time=31 ms

--- 2003::3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 15/25/32 ms

PC>
PC>
PC>ping 2002::2

Ping 2002::2: 32 data bytes, Press Ctrl_C to break
From 2002::2: bytes=32 seq=1 hop limit=63 time<1 ms
From 2002::2: bytes=32 seq=2 hop limit=63 time=31 ms
From 2002::2: bytes=32 seq=3 hop limit=63 time<1 ms
From 2002::2: bytes=32 seq=4 hop limit=63 time=16 ms
From 2002::2: bytes=32 seq=5 hop limit=63 time=16 ms
```

```
--- 2002::2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 0/12/31 ms

PC>
```

The IPv6 addresses of R2 and R3 can be pinged.

3.5 Checking IPv6 Routing Tables

- Check the IPv6 routing table of R1.

```
[R1]display ipv6 routing-table
Routing Table : Public
 Destinations : 8 Routes : 8

 Destination   : ::1                            PrefixLength : 128
 NextHop       : ::1                            Preference   : 0
 Cost          : 0                              Protocol     : Direct
 RelayNextHop : ::                              TunnelID     : 0x0
 Interface     : InLoopBack0                    Flags        : D

 Destination   : 2001::                         PrefixLength : 64
 NextHop       : 2001::1                        Preference   : 0
 Cost          : 0                              Protocol     : Direct
 RelayNextHop : ::                              TunnelID     : 0x0
 Interface     : GigabitEthernet0/0/2           Flags        : D

 Destination   : 2001::1                        PrefixLength : 128
 NextHop       : ::1                            Preference   : 0
 Cost          : 0                              Protocol     : Direct
 RelayNextHop : ::                              TunnelID     : 0x0
 Interface     : GigabitEthernet0/0/2           Flags        : D

 Destination   : 2002::                         PrefixLength : 64
 NextHop       : 2002::1                        Preference   : 0
 Cost          : 0                              Protocol     : Direct
 RelayNextHop : ::                              TunnelID     : 0x0
 Interface     : GigabitEthernet0/0/0           Flags        : D

 Destination   : 2002::1                        PrefixLength : 128
 NextHop       : ::1                            Preference   : 0
 Cost          : 0                              Protocol     : Direct
 RelayNextHop : ::                              TunnelID     : 0x0
 Interface     : GigabitEthernet0/0/0           Flags        : D

 Destination   : 2003::                         PrefixLength : 64
 NextHop       : 2002::2                        Preference   : 60
 Cost          : 0                              Protocol     : Static
 RelayNextHop : ::                              TunnelID     : 0x0
 Interface     : GigabitEthernet0/0/0           Flags        : RD
```

```
Destination   : 2005::                        PrefixLength : 64
NextHop       : 2002::2                       Preference   : 60
Cost          : 0                             Protocol     : Static
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : GigabitEthernet0/0/0          Flags        : RD

Destination   : FE80::                        PrefixLength : 10
NextHop       : ::                            Preference   : 0
Cost          : 0                             Protocol     : Direct
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : NULL0                         Flags        : D
```

- Check the IPv6 routing table of R2.

```
<R2>display ipv6 routing-table
Routing Table : Public
 Destinations : 8 Routes : 8

Destination   : ::1                           PrefixLength : 128
NextHop       : ::1                           Preference   : 0
Cost          : 0                             Protocol     : Direct
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : InLoopBack0                   Flags        : D

Destination   : 2001::                        PrefixLength : 64
NextHop       : 2002::1                       Preference   : 60
Cost          : 0                             Protocol     : Static
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : GigabitEthernet0/0/0          Flags        : RD

Destination   : 2002::                        PrefixLength : 64
NextHop       : 2002::2                       Preference   : 0
Cost          : 0                             Protocol     : Direct
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : GigabitEthernet0/0/0          Flags        : D

Destination   : 2002::2                       PrefixLength : 128
NextHop       : ::1                           Preference   : 0
Cost          : 0                             Protocol     : Direct
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : GigabitEthernet0/0/0          Flags        : D

Destination   : 2003::                        PrefixLength : 64
NextHop       : 2003::2                       Preference   : 0
Cost          : 0                             Protocol     : Direct
RelayNextHop : ::                             TunnelID     : 0x0
Interface     : GigabitEthernet0/0/1          Flags        : D

Destination   : 2003::2                       PrefixLength : 128
NextHop       : ::1                           Preference   : 0
Cost          : 0                             Protocol     : Direct
```

```
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : GigabitEthernet0/0/1  Flags        : D

Destination  : 2005::                PrefixLength : 64
NextHop      : 2003::3               Preference   : 60
Cost         : 0                     Protocol     : Static
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : GigabitEthernet0/0/1  Flags        : RD

Destination  : FE80::                PrefixLength : 10
NextHop      : ::                    Preference   : 0
Cost         : 0                     Protocol     : Direct
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : NULL0                 Flags        : D

<R2>
```

- Check the IPv6 routing table of R3.

```
<R3>display ipv6 routing-table
Routing Table : Public
 Destinations : 8 Routes : 8

Destination  : ::1                   PrefixLength : 128
NextHop      : ::1                   Preference   : 0
Cost         : 0                     Protocol     : Direct
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : InLoopBack0           Flags        : D

Destination  : 2001::                PrefixLength : 64
NextHop      : 2003::2               Preference   : 60
Cost         : 0                     Protocol     : Static
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : GigabitEthernet0/0/1  Flags        : RD

Destination  : 2002::                PrefixLength : 64
NextHop      : 2003::2               Preference   : 60
Cost         : 0                     Protocol     : Static
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : GigabitEthernet0/0/1  Flags        : RD

Destination  : 2003::                PrefixLength : 64
NextHop      : 2003::3               Preference   : 0
Cost         : 0                     Protocol     : Direct
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : GigabitEthernet0/0/1  Flags        : D

Destination  : 2003::3               PrefixLength : 128
NextHop      : ::1                   Preference   : 0
Cost         : 0                     Protocol     : Direct
RelayNextHop : ::                    TunnelID     : 0x0
Interface    : GigabitEthernet0/0/1  Flags        : D
```

```
Destination   : 2005::                          PrefixLength : 64
NextHop       : 2005::3                         Preference   : 0
Cost          : 0                               Protocol     : Direct
RelayNextHop  : ::                              TunnelID     : 0x0
Interface     : GigabitEthernet0/0/2            Flags        : D

Destination   : 2005::3                         PrefixLength : 128
NextHop       : ::1                             Preference   : 0
Cost          : 0                               Protocol     : Direct
RelayNextHop  : ::                              TunnelID     : 0x0
Interface     : GigabitEthernet0/0/2            Flags        : D

Destination   : FE80::                          PrefixLength : 10
NextHop       : ::                              Preference   : 0
Cost          : 0                               Protocol     : Direct
RelayNextHop  : ::                              TunnelID     : 0x0
Interface     : NULL0                           Flags        : D

<R3>
```

3.6 Configuring the DHCPv6 Server Function on R3

Enable the DHCPv6 server function on R3 and configure an IPv6 address for PC2. Create an IPv6 address pool, specify the prefix and prefix length of IPv6 addresses in the address pool, and configure IPv6 addresses that do not participate in automatic allocation (generally gateway addresses) and the IPv6 address of the DNS server.

- Configure DHCPv6 on R3.

```
[R3]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R3]
[R3]dhcpv6 pool pool1
[R3-dhcpv6-pool-pool1]address prefix 2005::/64
[R3-dhcpv6-pool-pool1]excluded-address 2005::3
[R3-dhcpv6-pool-pool1]quit
[R3]
```

- Configure the IPv6 address of G0/0/2 as the gateway address in the address pool. Configure the DHCPv6 server function and the address pool name.

```
[R3]int g0/0/2
[R3-GigabitEthernet0/0/2]dhcpv6 server pool1
[R3-GigabitEthernet0/0/2]quit
[R3]
```

3.7 Testing Network Connectivity on PC2

- Check the IPv6 address of PC2 obtained through DHCPv6.

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fe51:35d6
IPv6 address......................: 2005::1 / 128
```

```
IPv6 gateway......................: fe80::2e0:fcff:fe6d:72a8
IPv4 address......................: 0.0.0.0
Subnet mask.......................: 0.0.0.0
Gateway...........................: 0.0.0.0
Physical address..................: 54-89-98-51-35-D6
DNS server........................:

PC>
```

- Check the address allocation in the address pool on R3.

```
<R3>display dhcpv6 pool pool1
DHCPv6 pool: pool1
  Address prefix: 2005::/64
    Lifetime valid 172800 seconds, preferred 86400 seconds
    1 in use, 0 conflicts
  Excluded-address 2005::3
  1 excluded addresses
  Information refresh time: 86400
  Conflict-address expire-time: 172800
  Active normal clients: 1

<R3>

<R3>display dhcpv6 pool pool1 allocated address
Address                                    Valid     Expires              Left
-------------------------------------------------------------------------------
2005::1                                    172800    2025-04-06 15:10:04 172149
-------------------------------------------------------------------------------
Total : 1
<R3>
```

- On PC2, ping the IPv6 addresses of R2, R1, and PC1.

```
PC>ping 2001::10 -c 2

Ping 2001::10: 32 data bytes, Press Ctrl_C to break
From 2001::10: bytes=32 seq=1 hop limit=252 time=47 ms
From 2001::10: bytes=32 seq=2 hop limit=252 time=16 ms

--- 2001::10 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 16/31/47 ms

PC>ping 2002::1 -c 2

Ping 2002::1: 32 data bytes, Press Ctrl_C to break
From 2002::1: bytes=32 seq=1 hop limit=62 time=16 ms
From 2002::1: bytes=32 seq=2 hop limit=62 time=31 ms

--- 2002::1 ping statistics ---
```

```
   2 packet(s) transmitted
   2 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 16/23/31 ms

PC>ping 2003::2 -c 2

Ping 2003::2: 32 data bytes, Press Ctrl_C to break
From 2003::2: bytes=32 seq=1 hop limit=63 time=31 ms
From 2003::2: bytes=32 seq=2 hop limit=63 time=15 ms

--- 2003::2 ping statistics ---
   2 packet(s) transmitted
   2 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 15/23/31 ms

PC>
```

4 Verification

Verification has been performed during the lab.

5 Quiz

1. (Single-answer question) Which of the following statements about NDP-based stateless address autoconfiguration is correct? (D   )
   A. NDP-based stateless address autoconfiguration can be used to allocate 128-bit IPv6 addresses to hosts.
   B. NDP-based stateless address autoconfiguration can be used to allocate IPv6 prefixes or addresses to hosts.
   C. NDP-based stateless address autoconfiguration can be used to allocate only IPv6 prefixes to hosts.
   D. NDP-based stateless address autoconfiguration can be used to allocate DNS information to hosts.
2. (Multiple-answer question) Which of the following processes use Reply messages in DHCPv6? (ABCDE   )
   A. DHCPv6 information obtaining
   B. DHCPv6 prefix allocation's two-message exchange
   C. DHCPv6 address allocation's four-message exchange
   D. DHCPv6 address release
   E. DHCPv6 address lease renewal
3. What is the shortest form of the IPv6 address 2001:0DB8:0000:0000:032A:0000:0000:2D70?

   2001:DB8::32A:0:0:2D70 or 2001:DB8:0:0:32A::2D70